Robust Kernel Ridge Regression via Random Fourier Features

A Master's-Level Research Project Prospectus

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Abstract

This project studies kernel ridge regression when an adversary may replace an ϵ fraction of the training pairs. The proposed route first replaces a shift-invariant kernel by a finite random Fourier feature map and then applies robust finite-dimensional estimation. Random Fourier features are uniformly bounded, which removes some tail difficulties, but boundedness alone does *not* supply the relative moment conditions used by many sharp robust covariance results. Moreover, estimating a covariance matrix in operator norm does not justify whitening directions whose population variance lies below the estimation error. The project therefore asks whether ridge regularization, spectral truncation, and a source condition can make the unresolved directions irrelevant for prediction. This prospectus separates established ingredients from proposed claims, identifies the main mathematical gap, and gives a staged plan suitable for a master's student.

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1 Project overview

Kernel ridge regression (KRR) combines a flexible nonlinear model with a convex optimization problem. Its direct implementation, however, uses an $n \times n$ kernel matrix and is vulnerable to a small number of maliciously chosen training examples. Random Fourier features (RFFs) address the first issue by turning a shift-invariant kernel into an explicit m-dimensional feature map. This project asks whether the same reduction can also help with the second issue by converting robust KRR into robust ridge regression.

The intended pipeline is

KRR $\longrightarrow$ RFF ridge regression $\longrightarrow$ robust moment estimation $\longrightarrow$ regularized or truncated solution.

The central obstacle is spectral. The covariance of the RFF vector can have very small eigenvalues. An additive operator-norm estimate cannot resolve these directions, and naively inverting the estimated covariance amplifies the error. Prediction loss, on the other hand, downweights low-variance directions. This suggests that global conditioning may be unnecessary if the target has little prediction-relevant mass in the unresolved spectrum.

Primary objective. Prove and test a finite-dimensional result for robust ridge regression on RFF features that controls clean-population prediction error without assuming a uniform lower bound on the feature covariance.

Secondary objective. Determine when a standard source condition for the kernel integral operator implies the finite-dimensional spectral alignment needed by the RFF theorem.

Status convention. Throughout the document, **Established** denotes a standard fact or a claim proved in the text; **Proposed** denotes a plausible target whose precise assumptions and rate remain to be proved; and **Open** denotes a genuine technical question.

2 Statistical model and performance criterion

Let P be a distribution on $\mathcal{X} \times \mathbb{R}$. A clean pair $(X, Y) \sim P$ satisfies

$$Y = f_\star(X) + \xi, \qquad \mathbb{E}[\xi \mid X] = 0. \tag{1}$$

The noise assumptions will be specified for each theorem; useful starting points are bounded or conditionally sub-Gaussian noise. Later work may relax these to finite conditional moments using robust estimation of the feature–response cross-moment.

Definition 2.1 (Strong replacement contamination)**.** *Draw an i.i.d. clean sample* $S = ((X_i, Y_i))_{i=1}^n \sim P^n$*. After observing* S *and the realized random feature map, an adversary selects a set* $O \subseteq [n]$ *with* $|O| \le \epsilon n$ *and replaces every pair indexed by* O *arbitrarily. The learner observes only the resulting sample* $\widetilde{S}$*.*

This is stronger than a model in which only labels are corrupted. It also makes explicit that the RFF randomness is not hidden from the adversary.

The main loss is clean-population prediction error:

$$\|f - f_\star\|_{L^2(P_X)}^2 = \mathbb{E}_{X\sim P_X}\big[(f(X) - f_\star(X))^2\big]. \tag{2}$$

Coefficient error is deliberately secondary: it can be large in a direction that has negligible variance and hence negligible predictive importance.

3 Clean kernel ridge regression

Let k be a bounded positive-semidefinite kernel with reproducing kernel Hilbert space (RKHS) $\mathcal{H}_k$. Clean population ridge regression solves

$$f_\lambda = \arg\min_{f\in\mathcal{H}_k} \left\{\mathbb{E}(Y - f(X))^2 + \lambda \|f\|_{\mathcal{H}_k}^2\right\}, \tag{3}$$

while empirical KRR replaces the expectation by the sample average. By the representer theorem, the empirical minimizer has the form $\widehat{f}_\lambda(\cdot) = \sum_{i=1}^n \alpha_i k(X_i, \cdot)$, where, under the normalization in Equation (3),

$$\boldsymbol{\alpha} = (K + n\lambda I_n)^{-1}\boldsymbol{y}.$$

The population kernel integral operator $T : L^2(P_X) \to L^2(P_X)$ is

$$(Tg)(x) = \int k(x, x')g(x')\, dP_X(x'). \tag{4}$$

Two standard quantities governing clean KRR are the effective dimension

$$\mathcal{N}(\lambda) = \operatorname{tr}\big(T(T + \lambda I)^{-1}\big) \tag{5}$$

and a source condition such as

$$f_\star = T^r g, \qquad \|g\|_{L^2(P_X)} \le R. \tag{6}$$

The exact admissible range and interpretation of r depend on the convention and ambient space. The student should not import a rate from the literature without first matching these definitions. Caponnetto and De Vito [3] and the correction by Sutherland [10] are useful references for this operator viewpoint.

4 Random Fourier features

Suppose $\mathcal{X} \subseteq \mathbb{R}^d$ and $k(x, x') = \kappa(x - x')$ is continuous, shift invariant, and normalized so that $k(x, x) = 1$. Bochner's theorem implies that κ is the Fourier transform of a probability measure ρ:

$$\kappa(t) = \int_{\mathbb{R}^d} e^{i\omega^\top t}\, d\rho(\omega).$$

For the Gaussian kernel

$$k_\sigma(x, x') = \exp\left(-\frac{\|x - x'\|_2^2}{2\sigma^2}\right),$$

ρ is Gaussian with covariance $\sigma^{-2} I_d$. Draw $\omega_1, \dots, \omega_m \overset{\text{iid}}{\sim} \rho$ and $b_1, \dots, b_m \overset{\text{iid}}{\sim} \text{Unif}[0, 2\pi]$, and define

$$\Phi(x) = \sqrt{\frac{2}{m}}\big(\cos(\omega_1^\top x + b_1), \dots, \cos(\omega_m^\top x + b_m)\big)^\top. \tag{7}$$

Then $\mathbb{E}_{\omega,b} \langle \Phi(x), \Phi(x') \rangle = k(x, x')$; see Rahimi and Recht [8]. Ridge regression in this feature space is

$$\widehat{\beta}_\lambda = \arg\min_{\beta \in \mathbb{R}^m} \left\{ \frac{1}{n} \sum_{i=1}^n (Y_i - \beta^\top \Phi(X_i))^2 + \lambda \|\beta\|_2^2 \right\}, \qquad \widehat{f}(x) = \widehat{\beta}_\lambda^\top \Phi(x). \tag{8}$$

4.1 What bounded features do and do not imply

Let $Z = \Phi(X)$. The chosen normalization gives, deterministically,

$$\|Z\|_2^2 = \frac{2}{m} \sum_{j=1}^m \cos^2(\omega_j^\top X + b_j) \le 2. \tag{9}$$

Consequently $|v^\top Z| \le \sqrt{2}\, \|v\|_2$ and $\mathbb{E}|v^\top Z|^4 \le 4 \|v\|_2^4$ for every v.

This is useful absolute control, but the following distinction is essential. Sharp covariance results under weak moments often assume a *relative* marginal moment condition

$$\big(\mathbb{E}|v^\top (Z - \mathbb{E}Z)|^4\big)^{1/4} \le L\big(\mathbb{E}|v^\top (Z - \mathbb{E}Z)|^2\big)^{1/2} \quad \text{for all } v. \tag{10}$$

Equation (9) does not imply (10) with a uniform L: the variance in the denominator may be arbitrarily small. Thus the statement "RFFs are bounded, therefore a relative operator-norm covariance theorem applies" is not valid without an additional argument or assumption.

Remark 4.1 (Centering and second moments)**.** *For regression, the normal matrix is usually the second moment $M = \mathbb{E}[ZZ^\top]$, not the centered covariance $\Sigma = \mathbb{E}[(Z - \mathbb{E}Z)(Z - \mathbb{E}Z)^\top]$. They agree only when $\mathbb{E}Z = 0$. One may include an unpenalized intercept and center the features robustly, or work directly with M. This prospectus uses M unless centering is explicit.*

5 The finite-dimensional ridge problem

Condition on the realized feature map and define

$$M = \mathbb{E}[ZZ^\top], \qquad c = \mathbb{E}[ZY], \qquad \beta_\lambda = (M + \lambda I)^{-1} c. \tag{11}$$

The corresponding population RFF predictor is $f_{m,\lambda}(x) = \beta_\lambda^\top \Phi(x)$. If robust estimators $\widehat{M}$ and $\widehat{c}$ are available, the plug-in estimator is

$$\widetilde{\beta}_\lambda = (\widehat{M} + \lambda I)^{-1} \widehat{c}. \tag{12}$$

The ridge shift already prevents literal inversion of tiny eigenvalues. Nevertheless, the basic resolvent identity shows where a potentially harmful $1/\lambda$ factor enters:

$$\begin{aligned} \widetilde{\beta}_\lambda - \beta_\lambda = {} & (\widehat{M} + \lambda I)^{-1} (\widehat{c} - c) \\ & + (\widehat{M} + \lambda I)^{-1} (M - \widehat{M}) \beta_\lambda. \end{aligned} \tag{13}$$

An absolute bound $\left\|\widehat{M} - M\right\|_{\mathrm{op}} \le \delta$ combined only with $\left\|(\widehat{M} + \lambda I)^{-1}\right\|_{\mathrm{op}} \le 1/\lambda$ yields a term of order δ/λ. The project aims to replace this worst-case Euclidean argument by one adapted to the prediction geometry.

For any coefficient error h,

$$\left\|h^\top \Phi(\cdot)\right\|^2_{L^2(P_X)} = h^\top M h = \left\|M^{1/2} h\right\|_2^2. \tag{14}$$

Thus the natural perturbation object is not merely $\left\|\widehat{M} - M\right\|_{\mathrm{op}}$, but the ridge-relative error

$$\Delta_\lambda = \left\|(M + \lambda I)^{-1/2}(\widehat{M} - M)(M + \lambda I)^{-1/2}\right\|_{\mathrm{op}}. \tag{15}$$

When $\Delta_\lambda < 1$, standard matrix perturbation arguments compare the true and estimated ridge geometries without requiring M to be invertible. Obtaining such a guarantee robustly is one of the main technical tasks.

6 Spectral resolution and source alignment

Let $M = \sum_{j=1}^m \mu_j u_j u_j^\top$, with $\mu_j \ge 0$. Suppose only that $\left\|\widehat{M} - M\right\|_{\mathrm{op}} \le \delta$. Eigenvalues far below δ and eigenspaces separated by gaps comparable to δ cannot be expected to be accurately recovered. Full whitening by $\widehat{M}^{-1/2}$ is therefore unstable. Ridge regularization or thresholding at a level $\tau \gtrsim \delta$ is the appropriate default.

Consider the finite-dimensional source condition

$$\beta = M^r v, \qquad \|v\|_2 \le R, \qquad r \ge 0. \tag{16}$$

This says that the target coefficient decreases along low-variance eigendirections. Let $P_{<\tau} = \sum_{\mu_j < \tau} u_j u_j^\top$.

Proposition 6.1 (Prediction cost of discarding the bottom spectrum)**.** *Under* (16)*,*

$$\left\|M^{1/2} P_{<\tau} \beta\right\|_2 \le R\tau^{r+1/2}. \tag{17}$$

Proof. Write $v = \sum_j v_j u_j$. Then

$$\left\|M^{1/2} P_{<\tau} \beta\right\|_2^2 = \sum_{\mu_j < \tau} \mu_j^{2r+1} v_j^2 \le \tau^{2r+1} \|v\|_2^2.$$

Taking square roots proves the claim. □

This elementary inequality is the main motivation for the project, but it is not yet a robust regression theorem. Such a theorem must also control the estimated top spectral subspace, the cross-moment c, sampling error, and the interaction between λ and τ.

7 Robust estimation ingredients

7.1 Estimating the feature second moment

The matrices $Z_i Z_i^\top$ satisfy $\left\|Z_i Z_i^\top\right\|_{\mathrm{op}} \le 2$ for clean observations. This boundedness gives standard concentration for the clean empirical second moment. Under strong contamination, however, the corrupted feature vectors need not be bounded: the adversary replaces the observed pairs and is not required to provide points that are consistent with $\Phi(x)$. A robust procedure must therefore reject, filter, truncate, or reweight arbitrary submitted vectors.

Results such as Abdalla and Zhivotovskiy [1] and Oliveira and Rico [7] provide strong operator-norm guarantees under adversarial contamination and L_p–L_2 marginal equivalence. Minsker and Wang [6] study adversarial contamination and heavy tails with adaptation to low-rank structure. Their assumptions and conclusions should be checked carefully against (10); none should be invoked solely from the absolute boundedness in (9).

Three defensible routes are available:

(a) assume a relative L_4–L_2 condition for the RFF distribution and apply an existing robust covariance theorem;

(b) exploit the special trigonometric structure of RFFs to prove an appropriate ridge-relative estimate directly;

(c) settle initially for an absolute operator-norm bound, then combine it with thresholding and Equation (17).

Route (c) is the most suitable first master's-level target.

7.2 Estimating the cross-moment

Covariance estimation alone is insufficient. The normal equations also use $c = \mathbb{E}[ZY]$. Although Z is bounded, ZY inherits the tails of Y. Under bounded Y it is a bounded-vector robust mean problem. Under sub-Gaussian or finite-moment noise it requires an estimator and theorem matched to those assumptions. An alternative is to use a robust regression method directly rather than separately estimating M and c; robust linear regression results for Gaussian covariates illustrate what is possible in a more structured setting [5].

8 A precise target theorem template

The following is deliberately a template, not a claim.

Research question 8.1 (Finite-dimensional robust RFF ridge theorem)**.** *Under explicit assumptions on the conditional noise and the clean distribution of $Z = \Phi(X)$, construct a polynomial-time estimator $\widetilde{\beta}$ such that, with probability at least $1 - \eta$,*

$$\left\|M^{1/2}(\widetilde{\beta} - \beta_\star)\right\|_2 \le \mathcal{E}_{\mathrm{sample}}(n, m, \eta) + \mathcal{E}_{\mathrm{contam}}(\epsilon) + \mathcal{E}_{\mathrm{ridge}}(\lambda) + R\tau^{r+1/2}, \tag{18}$$

where τ is a justified robust spectral-resolution scale and no positive lower bound on $\lambda_{\min}(M)$ is assumed.

The theorem must state whether $\beta_\star$ is an unregularized population coefficient, the best linear predictor in the RFF class, or the ridge target β_λ. These are different objects. A clean first result should compare with β_λ; approximation to $f_\star$ can then be handled in a separate step.

8.1 A candidate truncated estimator

A concrete starting algorithm is:

1. Compute robust estimates $\widehat{M}$ of M and $\widehat{c}$ of c.
2. Diagonalize $\widehat{M} = \sum_j \widehat{\mu}_j \widehat{u}_j \widehat{u}_j^\top$ and form $\widehat{P}_\tau = \sum_{\widehat{\mu}_j \geq \tau} \widehat{u}_j \widehat{u}_j^\top$.
3. Return

$$\widetilde{\beta}_{\lambda,\tau} = \widehat{P}_\tau (\widehat{P}_\tau \widehat{M} \widehat{P}_\tau + \lambda \widehat{P}_\tau)^\dagger \widehat{P}_\tau \widehat{c}. \tag{19}$$

Here † denotes the Moore–Penrose pseudoinverse on the retained subspace. The analysis should avoid requiring individual eigenvectors to be identified when eigenvalues are clustered. Projector or functional-calculus arguments are preferable.

9 From the finite-dimensional result back to KRR

Let $\widehat{f}_{\mathrm{rob}}$ be the robust RFF estimator, $f_{m,\lambda}$ the population or clean empirical RFF ridge comparator (the choice must be fixed), and f_λ the corresponding exact-kernel ridge comparator. Then

$$\begin{aligned} \left\| \widehat{f}_{\mathrm{rob}} - f_\star \right\|_{L^2(P_X)} \leq & \underbrace{\left\| \widehat{f}_{\mathrm{rob}} - f_{m,\lambda} \right\|_{L^2(P_X)}}_{\text{robust finite-dimensional estimation}} \\ & + \underbrace{\| f_{m,\lambda} - f_\lambda \|_{L^2(P_X)}}_{\text{random-feature approximation}} \\ & + \underbrace{\| f_\lambda - f_\star \|_{L^2(P_X)}}_{\text{clean ridge bias / statistical error}} . \end{aligned} \tag{20}$$

This inequality is elementary, but each comparator must be defined on the same population/sample probability space. Mixing a population RFF solution with a finite-sample exact KRR solution obscures which randomness each term contains.

Rahimi and Recht [8] establish the RFF construction. Rudi and Rosasco [9] show that random-feature ridge regression can attain optimal learning behavior with substantially fewer features than samples in standard regimes. Avron et al. [2] study spectral approximation for KRR and explain the role of leverage-based feature sampling. These works provide tools for the middle term of (20), but the student must match their regularization conventions and hypotheses to the robust theorem.

10 The kernel-to-RFF source-condition question

The finite-dimensional assumption $\beta = M^r v$ is mathematically convenient but model-dependent: both M and the coefficient vector depend on the realized features. The conceptually cleaner assumption is the kernel-level condition $f_\star = T^r g$ in (6).

Research question 10.1 (Transfer of source regularity)**.** *For which feature-sampling schemes, feature counts m, and regularization levels λ does the kernel source condition imply either*

(i) a finite-dimensional source representation with controlled radius, or

(ii) directly a small prediction contribution from the empirically unresolved RFF spectrum?

Exact transfer may be too strong because the RFF subspace is random. A more realistic goal is an approximate statement with a residual term already controlled by the RFF approximation error. This question should be attempted only after the finite-dimensional analysis is complete.

11 Recommended project stages

Milestone 11.1 (Definitions and sanity checks; weeks 1–2)**.** *Fix all target objects, distinguish covariance from second moment, derive the normal equations and* (13)*, and reproduce Equation* (17)*. Construct a two-dimensional example showing why an additive covariance estimate cannot safely whiten a tiny eigenvalue.*

Milestone 11.2 (Clean RFF baseline; weeks 3–4)**.** *Implement exact KRR and RFF ridge regression for synthetic data. Empirically study how prediction error changes with m and λ. Read [8, 9, 2] and state one applicable approximation result using the project's notation.*

Milestone 11.3 (Robust finite-dimensional baseline; weeks 5–7)**.** *Begin with bounded responses and implement a robust mean estimator for c and a robust estimator for M. Compare a plug-in ridge estimator, full whitening, and spectral truncation under pair contamination.*

Milestone 11.4 (Perturbation theorem; weeks 8–10)**.** *Prove a deterministic theorem: conditional on bounds for $\widehat{M} - M$ and $\widehat{c} - c$, control prediction error of* (19) *under* (16)*. Keep the stochastic robust-estimation result as a separate corollary.*

Milestone 11.5 (Integration or counterexample; weeks 11–12)**.** *Combine the deterministic result with one carefully verified robust estimation theorem. If the desired rate cannot follow from the available assumptions, give a counterexample identifying the missing condition. Either outcome is a successful research contribution.*

Milestone 11.6 (Optional extension; weeks 13–14)**.** *Investigate transfer from* (6)*, leverage-weighted features, or noise weaker than boundedness. This is a stretch goal, not a prerequisite for completing the project.*

12 Experiments

Experiments should diagnose theory rather than merely compare test accuracy. A useful synthetic design chooses a prescribed diagonal feature covariance, places the target coefficient at controlled spectral locations, and corrupts an ϵ fraction of complete pairs. Record:

- clean prediction error versus ϵ, m, n, λ, and truncation threshold τ;
- operator error $\left\|\widehat{M} - M\right\|_{\mathrm{op}}$ and ridge-relative error Δ_λ;
- the target energy $\sum_{\mu_j < \tau} \mu_j \beta_j^2$ below the threshold;
- sensitivity to an adversary concentrated in a low-variance direction;
- comparison with nonrobust RFF ridge, oracle clean RFF ridge, and exact clean KRR.

The oracle comparators are essential: they distinguish kernel approximation, ordinary sampling error, and contamination error.

13 Common pitfalls

1. **Bounded is not relatively hypercontractive.** Do not infer (10) from (9).
2. **Covariance is not the regression normal matrix.** Account for the feature mean or use $M = \mathbb{E}[ZZ^\top]$ directly.
3. **The response moment must be estimated.** A robust covariance estimate does not by itself produce a regression estimator.
4. **Ridge and truncation have different biases.** Track both λ and τ rather than merging them informally.
5. **Eigenspaces need gaps; spectral functions often do not.** Avoid claims about recovering individual eigenvectors unless a gap assumption is stated.
6. **A displayed desired rate is not a theorem.** Every final result must identify its estimator, probability, sample-size conditions, constants or parameter dependence, and computational status.
7. **Condition on feature randomness consistently.** Say whether a statement holds for most feature draws, for a fixed draw, or jointly over feature and sample randomness.

14 Concrete deliverables

At the end of the project, the student should produce:

1. a precise problem statement and literature review;
2. a deterministic perturbation theorem for regularized or truncated moment-based regression;
3. either a stochastic corollary under a verified robust-estimation result or a counterexample showing why stronger assumptions are necessary;
4. reproducible synthetic experiments with oracle baselines;
5. a clear account of what remains open, especially the transfer from a kernel source condition to finite random features.

15 Questions to settle with the advisor

The following choices materially affect the theorem and should be fixed before the student begins the main proof.

1. Is the first theorem allowed to assume bounded responses, or should it begin immediately with conditionally sub-Gaussian noise?
2. Should the primary comparator be the population RFF ridge solution β_λ or the clean empirical RFF estimator?
3. Is polynomial-time computation required, or is an information-theoretic estimator acceptable for the first result?

4. Should complete-pair replacement remain the main model, or should the project first solve the simpler label-only contamination problem?

5. Is the kernel-level source-condition transfer a required deliverable or explicitly a stretch goal?

16 Reading guide

The following order is reasonable for a master's student.

1. Read Rahimi and Recht [8] for Bochner's theorem and the basic RFF construction.

2. Read selected sections of Rudi and Rosasco [9] for learning with random features, then Avron et al. [2] for the regularized spectral-approximation viewpoint.

3. Review KRR source conditions and effective dimension in Caponnetto and De Vito [3], together with Sutherland's correction [10].

4. Read a robust statistics overview such as Diakonikolas and Kane [4] to learn the strong contamination model and filtering paradigm.

5. Only then study robust covariance papers [1, 7, 6], writing down precisely which moment assumption each theorem uses.

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